

This question paper contains 8 printed pages]

13967

Roll No.

--	--	--	--	--	--	--	--	--	--	--	--

S. No. of Question Paper : 8079

Unique Paper Code : 2343012005

Name of the Paper : Data Mining-I

Name of the Course : B.Sc. (Hons.) Computer Science

Semester : IV

Duration : 3 Hours

Maximum Marks : 90

(Write your Roll No. on the top immediately on receipt of this question paper.)

The paper has two sections. Section A is compulsory.

Attempt any *four* questions from Section B.

Parts of the question must be answered together.

The use of simple calculator is allowed.

Section A

1. (a) What is class imbalance ? Why accuracy is not the right evaluation measure for classification of datasets having class imbalance ? 2
- (b) What are the *two* major categories of data mining tasks ? Give examples of each. 3
- (c) Why is sampling needed in data mining ? Name any *two* sampling methods. 3
- (d) What is the purpose of dimensionality reduction techniques ? 3
- (e) State the key differences between lazy learners and eager learners. Give an example of each type of learner. 3

P.T.O.

- (f) In the following dataset, identify the types of attributes, as nominal, ordinal, interval or ratio :

ID	Age	Gender	Income	Education Level	Purchase History	Temperature (°C)	Customer Rating
1	25	Male	55000	Bachelor's	Laptop	22.5	4
2	34	Female	75000	Master's	Tablet	18.0	5
3	28	Male	62000	Bachelor's	Camera	15.5	3
4	40	Female	85000	PhD	TV	19.5	5
5	60	Male	42000	High School	Washing Machine	25.0	2

- (g) Standardize the salary attribute and explain the significance of standardization :

Customer ID	Salary (in \$)	Age
C1	30,000	51
C2	45,000	34
C3	55,000	44
C4	65,000	43
C5	75,000	36

- (h) State any two methods for dealing with missing values. Explain each approach.
- (i) What are the estimates used for handling zero conditional probabilities ?

Section B

2. (a) Consider the following dataset containing information about customers and their income. The income attribute has been discretized into two different ways : Discretized Income Category I and Discretized Income Category II. Compute the entropy for these two discretized categories and determine which discretized category better helps in predicting whether the customer Purchased Product or not.

Customer ID	Income (in \$)	Discretized Income Category I	Discretized Income Category II	Purchased Product (Target)
1	20,000	Low	Lower	No
2	35,000	Medium	Lower	Yes
3	50,000	Medium	Upper	Yes
4	75,000	High	Upper	Yes
5	95,000	High	Upper	Yes
6	30,000	Medium	Lower	No
7	40,000	Medium	Lower	Yes
8	60,000	High	Upper	No
9	15,000	Low	Lower	No
10	55,000	Medium	Upper	Yes

Some logarithmic values are provided below :

$$\log_2(0.333) = -1.5864, \log_2(0.4) = -1.322, \log_2(0.5) = -1.0,$$

$$\log_2(0.6) = -0.737, \log_2(0.666) = -0.5864, \log_2(0.75) = -0.415.$$

(b) Identify whether the following represent classification or regression process ? 5

- (i) Predicting the price of a stock based on historical data.
 - (ii) Predicting whether a credit card transaction is fraudulent or legitimate.
 - (iii) Predicting the sales of a product for the next month or quarter.
 - (iv) Predicting whether a customer will purchase an item or not.
 - (v) Predicting the price of a house based on features such as size, location, number of rooms, etc.
- (c) Consider the following confusion matrix for a learning model to predict whether a patient is at risk of developing a particular disease : 5

		Predicted Disease	
		Positive	Negative
Actual	Positive	40	10
Disease	Negative	5	45

Evaluate the performance of the classifier using the following evaluation metrics :

Accuracy, Precision, Recall, True Positive Rate and True Negative Rate.

3. (a) Consider the following dataset, where each email is represented by features such as the presence or absence of certain words : 12

Email ID	Contains "Free"	Contains "Win"	Contains "Money"	Contains "Offer"	Label (Spam/Not Spam)
1	Yes	Yes	No	Yes	Spam
2	Yes	No	Yes	Yes	Spam
3	No	No	No	No	Not Spam
4	Yes	Yes	Yes	No	Spam
5	No	Yes	No	No	Not Spam
6	Yes	Yes	Yes	Yes	Spam
7	No	No	No	No	Not Spam
8	Yes	No	No	Yes	Spam
9	No	No	Yes	No	Not Spam
10	Yes	Yes	No	Yes	Spam

Use the Naive Bayes classifier to predict if a new email is spam or not based on the following features :

Contains "Free"=Yes, Contains "Win"=No, Contains "Money"=Yes", Contains "Offer"=No

- (b) How does the presence of missing values affect the performance of the Naive Bayes classifier ? 3

4. Consider a transaction dataset as shown in the following table :

Transaction ID	Items Purchased
T1	{A, B, D}
T2	{A, B, C, D}
T3	{A, B, C, D, E}
T4	{B, C}
T5	{A, B, D, F}
T6	{E, F}
T7	{A, B, F}
T8	{C, E}

- (a) Given minimum support as 40% apply the Apriori Algorithm to generate frequent itemsets. 10
- (b) Generate all association rules for the frequent itemsets. Compute the confidence of each generated rule. Taking the minimum confidence as 60%, final all strong association rules. 5
5. (a) Consider the following sample dataset and apply decision tree classifier. Calculate the gain in the Gini index when splitting on Gender, Income Level and Previous Purchase. Which attribute would the decision tree induction algorithm choose ? Justify your answer. 12

Customer ID	Gender	Income Level	Previous Purchase	Purchase (Target)
1	Male	Low	Yes	Yes
2	Female	High	No	No
3	Female	Medium	Yes	Yes
4	Male	Medium	No	No
5	Female	High	Yes	Yes
6	Male	Low	Yes	No
7	Female	Medium	No	No
8	Male	High	Yes	Yes
9	Female	Low	No	No
10	Male	Medium	Yes	Yes

- (b) How k -fold cross validation method is used for model evaluation ? 3

6. (a) Consider the following sample dataset. Using the K-Nearest Neighbors (KNN) algorithm, classify an Iris flower with attributes, Sepal Length = 5.4 cm and Petal Length = 3.7 cm. Use Manhattan distance metric. 10
- (i) consider $k = 3$ for classification
- (ii) consider $k = 6$ for classification

Sepal Length (cm)	Petal Length (cm)	Species (Target)
5.1	1.4	Setosa
4.9	1.4	Setosa
4.7	1.3	Setosa
7.0	4.7	Versicolor
6.4	4.5	Versicolor
6.9	4.9	Versicolor
6.3	6.0	Virginica
5.8	5.1	Virginica
7.1	5.9	Virginica
6.3	5.1	Virginica

- (b) What are outliers in a dataset ? Why is it important to detect and analyze them in data mining ? Are all noise points considered outliers ?

7. (a) Consider a dataset for K-means clustering. Perform two iterations of the K-means clustering algorithm and show the clusters, their new centroids, and compute SSE after each iteration. Use Euclidean distance as proximity measure and the following three as Initial centroids : 12

Centroid 1 : Observation 1 $\rightarrow$ (4, 1, 1)

Centroid 2 : Observation 2 $\rightarrow$ (2, 3, 1)

Centroid 3 : Observation 3 $\rightarrow$ (1, 2, 4)

Observation ID	X	Y	Z
1	4	1	1
2	2	3	1
3	1	2	4
4	5	2	1
5	3	5	2
6	1	3	5

- (b) Describe well-separated clusters and density-based clusters, providing examples of each type. 3